

TROITSKOYE, Russian Federation

WMO: 316550

Lat: 49.450N

Lon: 136.567E

Elev: 122

StdP: 14.63

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.6	-20.7	-29.6	1.0	-23.9	-25.6	1.3	-19.5	23.4	-1.9	21.5	-2.4	2.3	180	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	85.4	72.5	82.4	71.2	79.8	69.5	74.9	82.1	73.0	79.8	71.3	77.7	5.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	121.0	79.5	70.7	113.7	77.4	69.1	107.3	75.6	38.6	82.2	36.9	80.0	35.3	77.9	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.5	18.1	15.6	-31.9	89.8	6.1	3.0	-36.2	91.9	-39.8	93.7	-43.2	95.3	-47.6	97.5		
			-32.0	78.1	6.1	2.4	-36.4	79.9	-39.9	81.3	-43.3	82.7	-47.7	84.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.7	-5.6	2.5	18.6	38.8	54.0	64.5	71.3	67.9	56.8	40.5	18.1	-1.6
	DBStd	28.27	9.66	9.41	9.82	8.87	7.71	6.59	5.40	4.97	6.94	8.15	11.34	8.97
	HDD50	7318	1725	1330	974	356	47	1	0	0	20	310	956	1599
	HDD65	11127	2190	1750	1439	788	348	88	13	27	256	758	1406	2064
	CDD50	2084	0	0	0	20	172	436	660	556	224	16	0	0
	CDD65	416	0	0	0	1	7	73	208	118	9	0	0	0
	CDH74	2544	0	0	0	6	93	510	1367	535	33	0	0	0
	CDH80	561	0	0	0	0	14	107	370	69	1	0	0	0
(14) Wind	WSAvg	6.3	6.1	6.0	6.1	6.1	6.7	5.8	5.6	5.4	6.3	6.7	7.7	7.3
(15) Precipitation	PrecAvg	25.4	0.6	0.7	1.0	1.5	2.4	2.6	4.0	4.6	3.0	1.7	1.3	1.1
	PrecMax	34.0	1.7	2.5	2.4	3.9	5.0	5.1	8.6	7.8	6.5	3.8	3.6	3.6
	PrecMin	19.1	0.1	0.0	0.0	0.2	0.7	0.5	0.0	2.2	1.0	0.2	0.2	0.2
	PrecStd	4.4	0.4	0.5	0.7	0.8	1.2	1.3	2.2	1.7	1.4	0.7	0.8	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.3	32.0	46.0	73.0	82.3	87.2	90.6	85.0	77.6	68.1	47.7	26.4
		MCWB	20.6	27.1	37.5	55.3	64.0	72.8	73.5	74.8	67.3	56.5	41.0	25.1
	2%	DB	15.9	26.3	40.7	64.8	76.5	83.1	87.0	81.8	74.2	61.5	41.8	19.3
		MCWB	14.3	23.1	34.6	51.0	60.6	69.9	73.9	72.1	65.2	52.3	36.5	17.9
	5%	DB	11.6	21.4	36.6	58.4	72.4	79.8	84.1	79.3	70.8	56.3	37.9	14.9
		MCWB	10.4	18.6	31.5	47.2	58.3	67.4	72.5	70.6	62.9	48.7	34.1	13.3
	10%	DB	8.1	17.2	33.3	53.0	68.7	76.4	81.1	77.0	67.7	52.5	33.9	11.4
		MCWB	7.0	15.2	29.5	44.2	56.5	65.9	71.0	69.3	61.0	46.3	30.7	10.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.9	27.0	38.6	57.1	67.4	75.9	77.5	76.8	69.0	58.0	41.8	25.1
		MCDB	22.0	30.5	45.2	70.4	75.0	83.2	86.2	82.7	75.5	66.4	47.2	26.5
	2%	WB	14.5	22.9	34.9	52.2	62.8	71.8	75.6	73.8	66.4	53.4	37.1	18.0
		MCDB	15.7	26.3	40.0	62.9	73.7	80.8	84.0	79.5	72.4	59.6	41.1	19.3
	5%	WB	10.5	18.7	32.1	48.5	60.0	69.0	73.9	72.0	64.2	50.1	34.3	13.5
		MCDB	11.6	21.2	36.0	57.5	69.8	77.4	81.3	77.6	69.6	55.0	37.5	14.8
	10%	WB	6.9	15.1	29.5	44.4	57.3	66.9	72.3	70.4	62.1	46.8	31.3	10.2
		MCDB	8.0	17.1	33.2	52.6	66.9	74.6	79.2	75.4	66.5	51.7	33.8	11.4
(35) Mean Daily Temperature Range	5% DB	MDBR	13.8	16.5	17.2	17.0	18.0	16.3	14.9	13.7	15.8	15.0	12.3	11.9
		MCDBR	14.1	18.0	18.0	23.9	25.1	21.2	19.5	17.5	19.3	20.3	14.5	13.7
	5% WB	MCWBR	12.8	15.4	13.3	13.9	13.8	10.5	9.2	9.2	11.2	13.1	10.6	12.6
		MCWBR	13.7	17.8	17.0	21.7	22.4	18.7	17.0	14.8	16.6	18.7	13.5	13.4
(40) Clear-Sky Solar Irradiance	taub		0.285	0.332	0.401	0.521	0.565	0.504	0.500	0.438	0.402	0.434	0.370	0.295
		taud	2.075	1.949	1.838	1.766	1.769	2.024	2.080	2.266	2.280	1.996	2.000	2.054
	Ebn at Noon		232	246	246	229	226	244	242	253	250	206	190	206
		Edh at Noon	27	40	54	65	67	52	49	39	35	38	30	24
(44) All-Sky Solar Radiation	RadAvg		538	924	1385	1515	1670	1820	1691	1452	1143	728	449	400
	RadStd		22	29	75	148	132	201	144	97	78	54	45	27

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page